

**Q-Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.**

DNS Troubleshooting : musicmania.in Resolution Failure

A identify and resolve the issue causing the DNS server to fail when resolving the domain **musicmania.in**. The troubleshooting process includes verifying DNS configuration, restarting the DNS service, and testing name resolution using DNS utilities.

Step 1: Verify DNS Configuration Files

Check the DNS configuration to ensure the zone and records are correctly configured.

For BIND DNS Server

Check the main configuration file:

sudo nano /etc/bind/named.conf.local

Verify that the zone entry exists:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

Next, verify the zone file:

sudo nano /etc/bind/db.musicmania.in

Ensure the **SOA**, **NS**, and **A** records are correct.

Example:

```
@    IN  SOA ns1.musicmania.in. admin.musicmania.in. (
      2026080401
      3600
      1800
      604800
      86400 )

@      IN  NS      ns1.musicmania.in.
ns1    IN  A       192.168.1.10
@      IN  A       192.168.1.20
www    IN  A       192.168.1.20
```

Step 2: Check DNS Configuration Syntax

Before restarting the DNS service, validate the configuration.

Check overall configuration:

```
sudo named-checkconf
```

Check the zone file:

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

Expected output:

OK

If errors are displayed, correct them before proceeding.

Step 3: Restart the DNS Service

After correcting any configuration issues, restart the DNS server.

For BIND:

```
sudo systemctl restart bind9
```

Check whether the service is running:

```
sudo systemctl status bind9
```

Expected output:

Active: active (running)

Step 4: Test DNS Resolution

Use **nslookup** to verify that the domain resolves correctly.

```
nslookup musicmania.in
```

Expected output:

Server: 192.168.1.10

Address: 192.168.1.10#53

Name: musicmania.in

Address: 192.168.1.20

Alternatively, use **dig**:

```
dig musicmania.in
```

Verify that the **ANSWER SECTION** contains the correct IP address.

Step 5: Check DNS Logs

Review DNS logs for configuration or runtime errors.

```
sudo journalctl -u bind9
```

or

```
sudo tail -f /var/log/syslog
```

Look for messages such as:

- Zone loading errors
 - Permission denied
 - Syntax errors
 - Failed queries
-

Step 6: Verify Firewall Settings

Ensure DNS traffic on port **53** is allowed.

For UFW:

```
sudo ufw status
```

Allow DNS if necessary:

```
sudo ufw allow 53
```

Step 7: Check Network Connectivity

Verify that the DNS server is reachable.

```
ping 192.168.1.10
```

Also verify that port 53 is listening:

```
sudo ss -tulnp | grep :53
```

Troubleshooting Commands Used

The following commands were used during troubleshooting:

1. `named-checkconf`
2. `named-checkzone musicmania.in /etc/bind/db.musicmania.in`
3. `systemctl restart bind9`
4. `systemctl status bind9`
5. `nslookup musicmania.in`
6. `dig musicmania.in`
7. `journalctl -u bind9`
8. `ss -tulnp | grep :53`

The DNS resolution issue for **musicmania.in** was investigated by checking the DNS configuration files, validating the zone configuration, restarting the DNS service, and testing resolution using **nslookup** and **dig**. Additional checks included reviewing system logs, verifying firewall rules, and confirming that the DNS service was listening on port 53. Following these troubleshooting steps helps identify configuration errors and ensures the DNS server resolves the domain correctly.